COMPUTATION OF THE ALTERNATING PRODUCT OF THE CHERN CLASSES OF THE EXTERIOR POWERS OF A LOCALLY FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FEE (sources/stacks-chow/pinned-original-source.tex line 12500).

Allowed standard prerequisites (editorial).

- The splitting principle for Chern classes of a locally free module and Chern roots. Same chapter, chow.tex, Section labelled section-splitting-principle (source line 8422), invoked at proof lines 12512-12514. Justifies replacing the computation on X by a computation in the polynomial ring on the Chern roots x_1, ..., x_r and supplies the formula c(wedge^n E) as a product over increasing multi-indices.
- The base situation (S, delta) for bivariant operational Chow rings. Same chapter, Situation labelled situation-setup (source line 1372), invoked in the statement. Fixes the ring A^*(X) in which the identity is stated.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UK: sources/stacks-chow/pinned-original-source.tex, source lines 8422–8422. Same-chapter pointer into the author environment lemma at source line 8422; invoked at proof line 12514 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of rank r. Then*

$$\prod\nolimits_{n=0,\ldots,r} c(\wedge^n \mathcal{E})^{(-1)^n} = 1 - (r-1)!c_r(\mathcal{E}) + \ldots$$

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Proof. By the splitting principle we can turn this into a calculation in the polynomial ring on the Chern roots $x_1, \ldots, x_r$ of $\mathcal{E}$. See Section 02UK. Observe that

$$c(\wedge^n \mathcal{E}) = \prod\nolimits_{1 \leq i_1 < \ldots < i_n \leq r} (1 + x_{i_1} + \ldots + x_{i_n})$$

Thus the logarithm of the left hand side of the equation in the lemma is

$$- \sum\nolimits_{p \geq 1} \sum\nolimits_{n=0}^{r} \sum\nolimits_{1 \leq i_1 < \ldots < i_n \leq r} \frac{(-1)^{p+n}}{p} (x_{i_1} + \ldots + x_{i_n})^p$$

Please notice the minus sign in front. However, we have

$$\sum\nolimits_{p \geq 0} \sum\nolimits_{n=0}^{r} \sum\nolimits_{1 \leq i_1 < \ldots < i_n \leq r} \frac{(-1)^{p+n}}{p!} (x_{i_1} + \ldots + x_{i_n})^p = \prod (1 - e^{-x_i})$$

Hence we see that the first nonzero term in our Chern class is in degree r and equal to the predicted value. $\square$